Customer Churn Analytics

Subscriber retention and revenue exposure

An OTT subscription case study using SQL, Python and Power BI

CUSTOMER BASE	CHURN RATE	LOST MONTHLY CHARGES
5,000	28.64%	17,707.68

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Project scope

An analysis of 5,000 OTT subscribers, examining cancellation patterns, monthly revenue exposure and support interactions to identify customer retention priorities.

Contents

SECTION	PAGE
Introduction	3
Data Exploration	4-5
Methods	6-7
Modelling and Results	8-11
Conclusion	12

Executive overview

The customer base recorded 1,432 cancellations, representing a churn rate of 28.64%. Canceled subscriptions account for 17,707.68 in monthly recurring charges. Retention efforts should focus on the customer groups contributing most to these losses.

The analysis integrates customer, subscription and support records into a single customer-level view. It examines commercial and service-related patterns, quantifies the concentration of churn and sets out actions for improving retention.

Retention priorities

Basic subscribers recorded a churn rate of 60.64%, compared with 14.58% for Premium subscribers. Monthly contracts recorded 55.15% churn, compared with 8.72% for annual contracts. These segments warrant targeted reviews of pricing, customer value and renewal experience, supported by controlled retention campaigns.

Introduction

The Business Challenge

OTT services compete for recurring subscriber spend across alternatives such as Netflix, Hotstar and Prime Video. Sustaining retention requires understanding which subscribers cancel, where losses concentrate and how service interactions relate to churn.

This project examines subscriber demographics, subscription tiers and support escalations to identify customer groups with elevated churn. The analysis connects cancellation patterns with recurring charges and service interactions to support retention decisions.

Business objectives

Measure churn and retention across the customer base; quantify monthly charges associated with canceled subscriptions; compare customer segments; assess the relationship between support escalations and cancellation; and recommend targeted retention actions.

Core Tech Stack

CAPABILITY	TOOLS AND APPLICATION
SQL and Python integration	SQLite, sqlite3 and pandas for relational extraction and analysis.
Data preparation and performance engineering	NumPy and pandas for data cleaning, feature engineering, quality checks and customer-level aggregation.
Behavioral visualization	Matplotlib and Seaborn for segment comparisons, time trends and exploratory correlations; Power BI for interactive reporting.

Definition of churn & churn rate

A subscriber is classified as churned when cancellation_date is populated. Subscribers without a cancellation date are classified as active. Churn rate is calculated as canceled subscribers divided by the total customer base; retention rate is the remaining proportion.

Data Exploration

Data structure and coverage

The customer_churn database contains three related tables: customer demographics, subscription records and support interactions. The customer and subscription tables each contain 5,000 distinct subscribers. The support table contains 5,000 interactions across 2,856 customers. SQL queries load these records into pandas DataFrames for preparation and analysis.

TABLE	ROWS	GRAIN
db_customer	5,000	One row per customer
db_subscription	5,000	One row per subscriber
db_support	5,000	One row per support interaction

Database Tables

The database uses customerid to connect subscriber profiles, commercial records and support history. The table structure is outlined below.

db_customer	db_subscription	db_support
customerid name country State gender dob interests pincode	customerid subscription_start_date subscription_type renewal_date plan_type contract_type cancellation_date cancellation_reason monthly_charges cltv churn_score	customerid complaint_date Escalations csat_score col_1 comment

Customer and subscription records have a one-to-one relationship. Support records have a one-to-many relationship with customers. Support interactions are therefore summarized at customer level before calculating subscriber metrics.

Data quality and exploration

OBSERVATION	DATA PREPARATION
3 missing country values	Filled missing country values using the state-to-country mapping.
Gender labels include Men and Women	Standardized labels to Male and Female.
17 missing interests and 21 missing pincodes	Excluded interests and pincode from the analytical feature set.
3,568 missing cancellation dates and reasons	Retained missing cancellation values to identify active subscriptions.
col_1 is entirely empty	Removed the empty column from the support table.
5 missing support comments	Excluded free-text comments and retained structured support indicators.
Repeated support customer IDs	Calculated complaint frequency and retained the latest interaction per customer.

Features selected for analysis

Plan type, contract type and acquisition channel define the principal commercial segments. Cancellation dates and monthly charges measure the timing and revenue exposure of churn. Geography supports regional comparisons, while escalation status, customer satisfaction and complaint frequency describe the service experience. Birth dates and subscription start dates support age and tenure calculations.

Customer and support coverage

Support records cover 57.12% of subscribers. The remaining 2,144 customers have no recorded support interaction and remain included in the customer base. Geographic analysis covers India and Nepal; the location field contains Indian states and Kathmandu.

Methods

Churn Analysis Roadmap

STAGE	IMPLEMENTATION
1) Connect to SQL	Connected to customer_churn.db with sqlite3 and identified the database tables through sqlite_master.
2) Import data	Executed SQL SELECT queries with pandas.read_sql to load each table into a DataFrame.
3) Clean data	Inspected data types and null values, renamed columns, selected relevant fields, standardized labels and converted date fields.
4) Engineer features	Created churn flags, complaint counts, tenure, age and risk bands using pandas and NumPy transformations.
5) Analyze behavior	Calculated customer and segment KPIs through aggregation, groupby operations and filtered analysis.
6) Visualize and communicate	Developed Matplotlib charts, Seaborn correlation and pair plots, and a Power BI dashboard to communicate retention patterns.

Building a customer-level analytical table

A direct join to support interactions produced 7,144 rows because individual customers could have multiple complaints. Complaint counts were calculated before retaining the latest dated support interaction for each customer. Left joins then combined subscriptions, demographics and support summaries into a 5,000-row analytical table with one record per subscriber.

The resulting table preserves the full subscriber base, including customers without support history. Each customer record combines subscription details, demographic attributes, complaint frequency and the most recent escalation and satisfaction indicators.

Feature and metric definitions

METRIC OR FEATURE	DEFINITION AND INTERPRETATION
Churn and retention	Churn flag = 1 when cancellation_date exists, otherwise 0. Churn rate = sum(flags) / 5,000. Retention = 1 - churn rate.
Segment churn rate	Canceled customers in a segment / all customers in that segment.
Lost monthly recurring charges	Sum of monthly_charges for canceled subscribers. Total: 17,707.68.
Average monthly charges	Mean monthly_charges across all 5,000 subscribers. Average: 18.83.
Complaint count	Number of support interactions per customer before retaining the latest record. Total interactions / total customers = 1.00.
Latest escalation prevalence	Customers whose latest support interaction has escalations = Y, divided by all 5,000 subscribers. Rate: 25.26%.
Tenure and age	Tenure uses cancellation date for canceled customers and execution date otherwise. Age uses execution date and days / 365, rounded.
Risk bands	Churn score bands: Low below 50; Medium from 50 to below 70; High at 70 or above.

Revenue measurement

Lost monthly recurring charges are calculated by filtering canceled subscriptions and summing monthly_charges. The resulting value of 17,707.68 measures the monthly charges associated with canceled accounts. Average monthly charges of 18.83 are calculated across the full customer base.

Financial values are expressed in currency units. Churn and retention percentages describe the customer base as a whole; monthly charts show cancellation counts grouped by calendar period.

Modelling and Results

Customer base and financial exposure

INDICATOR	RESULT
Total subscribers	5,000
Canceled subscribers	1,432
Subscribers without recorded cancellation	3,568
Churn rate / Retention rate	28.64% / 71.36%
Monthly charges associated with canceled subscribers	17,707.68 currency units
Average monthly charges across the full base	18.83 currency units
Average support interactions per subscriber	1.00
Latest escalation prevalence across all subscribers	25.26%

Plan-level churn

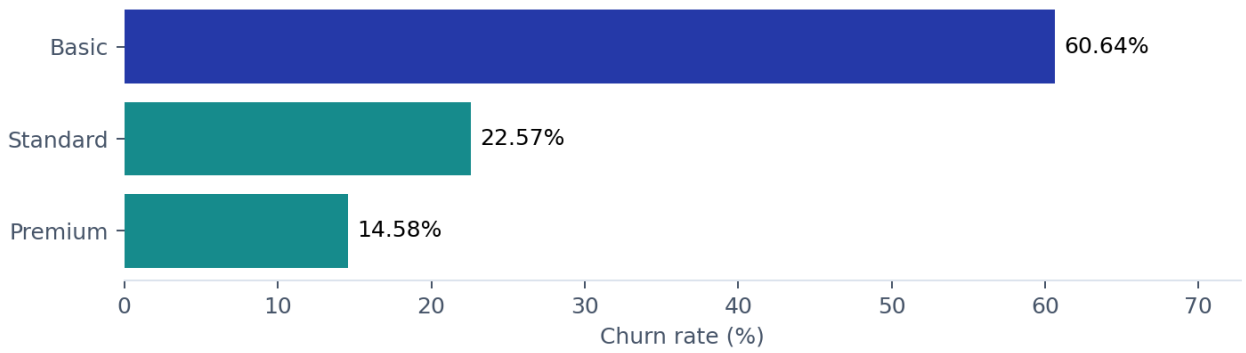


Figure 1. Churn rate by subscription plan.

Basic subscribers account for 701 cancellations among 1,156 customers, a churn rate of 60.64%. Standard records 482 cancellations among 2,136 customers (22.57%), while Premium records 249 among 1,708 (14.58%). Basic contributes 48.95% of all cancellations, making it a priority for reviewing the plan offering and customer experience.

The analysis uses customer segmentation and score-based risk bands to organize retention priorities. Plan-level findings should guide targeted campaigns alongside customer value, complaint history and cancellation reasons.

Contract and service patterns

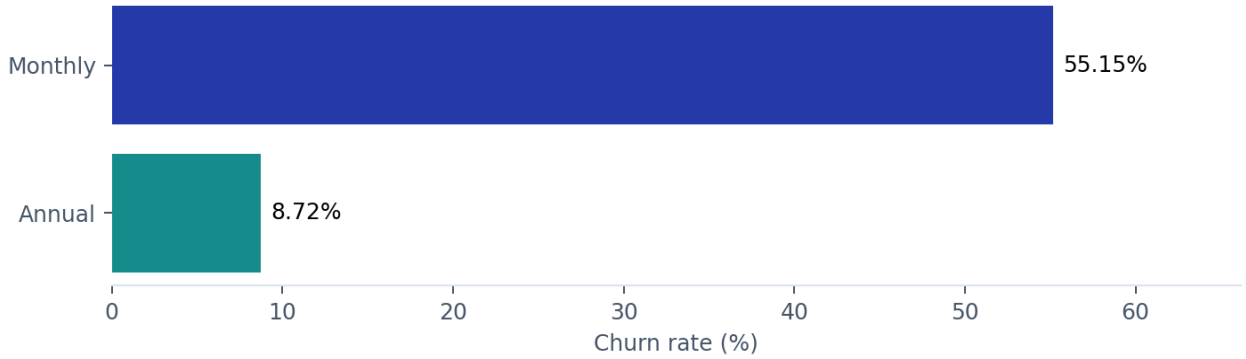


Figure 2. Churn rate by contract type.

Monthly contracts record 1,183 cancellations among 2,145 subscribers (55.15%). Annual contracts record 249 among 2,855 subscribers (8.72%). The monthly-contract churn rate is approximately 6.32 times the annual rate. A controlled annual-plan offer for eligible monthly subscribers would help assess the retention benefit and cost of contract migration.

Acquisition channels

CHANNEL	SUBSCRIBERS	CANCELED	CHURN
Organic	2,150	0	0.00%
Paid	1,449	249	17.18%
Referral	1,401	1,183	84.44%

Referral subscribers record the highest channel churn at 84.44%, followed by Paid at 17.18%. Organic subscribers record no cancellations. Review referral offer terms, onboarding and the mix of plans and contracts within each channel to understand this concentration.

Support escalation and churn

The latest support escalation indicator has a Pearson correlation of approximately 0.42 with churn. This positive association makes escalated complaints a useful focus for retention investigation. Customers whose latest interaction was escalated represent 25.26% of the subscriber base.

A service recovery process should prioritize active subscribers with unresolved or repeated complaints. Review complaint history, customer satisfaction and account value together when deciding which customers to contact and which issues require immediate attention.

Timing geography and reporting

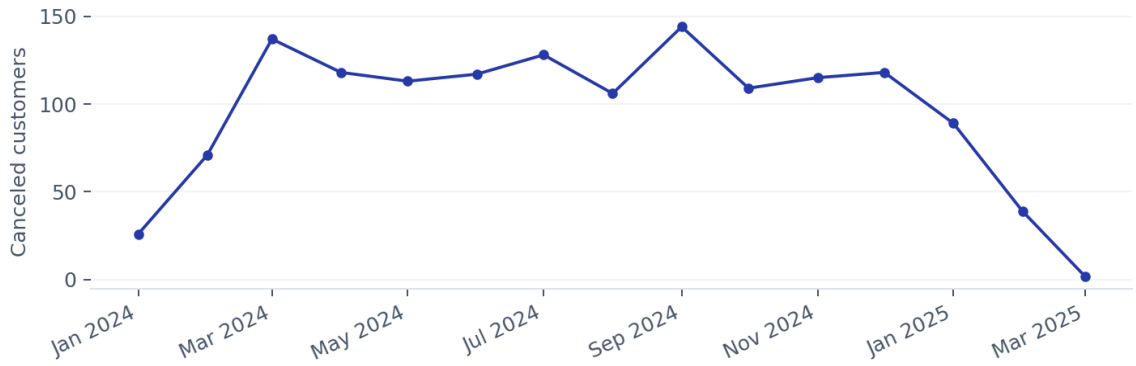
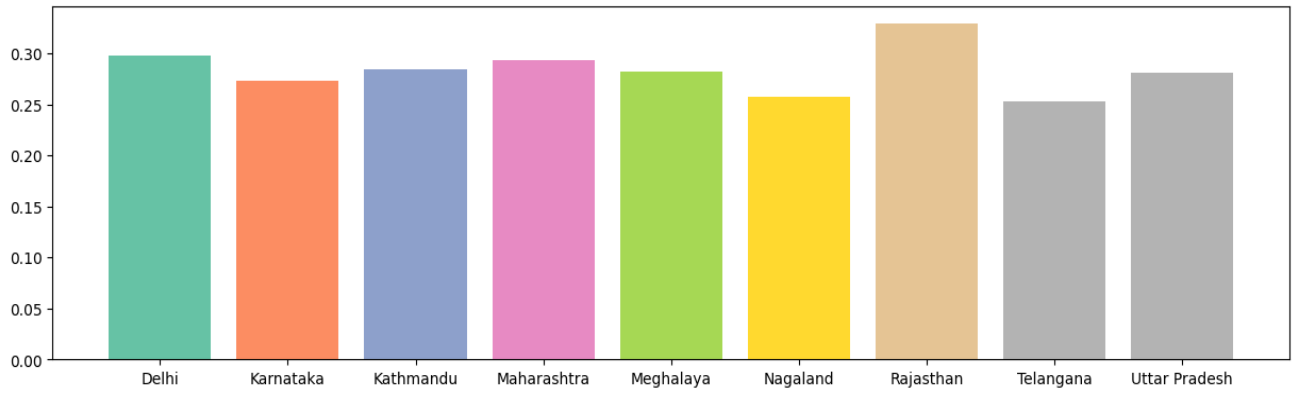


Figure 3. Monthly cancellation trend from January 2024 to March 2025.

Cancellations span January 2024 to March 2025. September 2024 records the highest monthly count at 144, associated with 1,729.56 in lost monthly charges. Review pricing changes, service incidents and campaign activity around this period to identify potential contributors to the peak.

Location findings

Rajasthan records the highest location churn rate at 32.92%, with 160 cancellations among 486 subscribers. Delhi records the largest cancellation volume at 281, representing 29.77% of its 944 subscribers. Telangana records 25.31%, with 123 cancellations among 486 subscribers. Delhi is a priority for reducing cancellation volume, while Rajasthan warrants investigation of its elevated churn rate.



Power BI Dashboard



Figure 4. Customer churn dashboard showing subscriber KPIs, plan and contract comparisons, cancellation trends, revenue exposure and geographic distribution.

Conclusion

The customer base records a churn rate of 28.64%, with cancellations concentrated in Basic plans and monthly contracts. These segments present the clearest priorities for retention activity. Referral acquisition also warrants review, while the relationship between escalation and churn highlights the importance of service recovery. Geographic targeting should address Delhi's cancellation volume and Rajasthan's elevated churn rate.

Recommended next steps

PRIORITY	PROPOSED ACTION	SUCCESS MEASURE
1	Investigate Basic and monthly subscriber cancellation reasons, pricing history and service experience.	Cohort churn and lost monthly charges, adjusted for observation period.
2	Test annual-plan offers among eligible active monthly subscribers using a control group.	Incremental retained MRR, conversion and margin after incentives.
3	Establish a service recovery queue for active customers with unresolved or repeated complaints, prioritized by customer value and risk.	Resolution time, post-contact cancellation and CSAT change.
4	Review Delhi cancellation volume and Rajasthan relative churn alongside country and cohort composition.	Location churn using consistent geography and comparable cohorts.
5	Evaluate churn-score performance and use pre-cancellation customer history to strengthen risk-based prioritization.	Risk-band cancellation rates, campaign response and incremental retention.

Business implications

A focused retention program should combine improvements to the Basic plan experience, targeted contract offers and timely complaint resolution. Customer-level reporting can track these interventions through churn, retained monthly revenue and customer satisfaction. Controlled campaign evaluation will establish which actions deliver sustainable improvements.